INCH-POUND

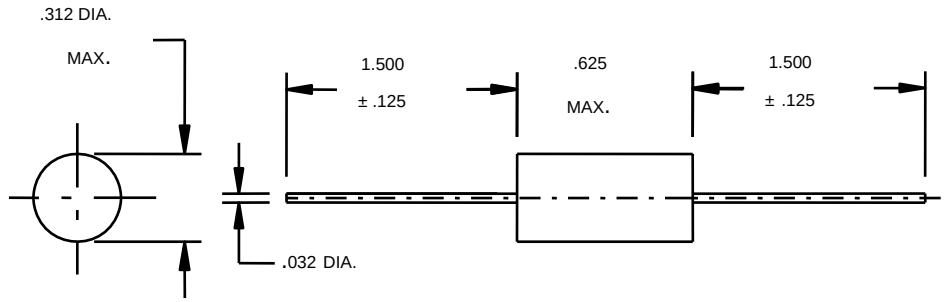
MIL-PRF-27/88A
29 September 2009
SUPERSEDING
MIL-PRF-27/88
22 April 1976

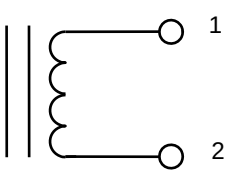
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER, AND HIGH-POWER PULSE),
INDUCTOR, POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.





72 uH MIN.
.1 V
10kHz
.3 OHMS MAX.
1.0 Adc

WV : 50 VOLTS MAX.

ALTITUDE : 70.000 FT. MAX.

Inches	mm
.032	0.81
.125	3.18
.312	7.92
.625	15.88
1.500	38.10

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses, such as (1.-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Inductance (1-2): 72µH minimum.

Voltage (1-2): 0.1 volts rms, 10 kHz.

Power: 0.3 watt.

Current (1-2): 1 ampere dc.

DC resistance(1-2): 0.3 ohm dc, maximum.

Working voltage: 50 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Altitude: 70,000 feet, maximum.

Terminals: Solid wire lead type, tinned soft copper.

Weight: 3.2 grams.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 100 volts rms.

At reduced barometric pressure: 50 volts rms.

Electrical characteristics:

Inductance: With 0.1 volt rms, 10 kHz across, and 1 ampere dc applied to (1-2) the inductance shall be 72 µH minimum.

DC resistance at 25°C: 0.3 ohm across (1-2) maximum.

Self resonant frequency: 2.5 MHz minimum.

Temperature rise: 30˚C maximum, with 0.1 volt rms, 10 kHz, 1 ampere dc applied to (1-2) at an ambient temperature of 100°C.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/88-01.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2009-033)

Review activities:
Army - AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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